

Computer Networks



Module 4: SNMP, ASN.1, Physical Layer

- SNMP
- ASN.1
- Physical Layer:
 - Data and signals
 - Digital transmission
 - Analog transmission
 - Bandwidth utilization
 - Transmission media

7.1 Data and Signals

Physical Layer Perspective

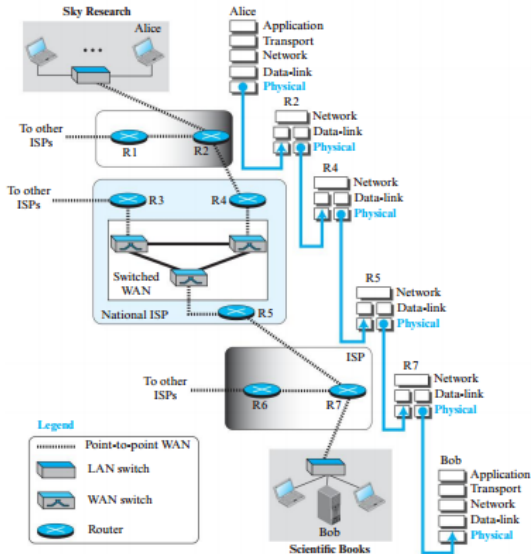
At the physical layer, communication takes place directly between two neighboring nodes. The nodes exchange information using **electromagnetic signals**.

- A computer stores information as bits: **0 and 1**.
- Bits cannot simply be placed directly onto a wire or into the air.
- The bits must first be converted into electromagnetic signals.
- The major responsibility of the physical layer is therefore to efficiently convert bits into signals and transmit them.

Key Idea

Bits are information; signals are the physical representation used to carry that information.

Figure 7.1 Communication at the physical layer



Analog and Digital Data

Analog Data

- Continuous in nature.
- Has continuously varying values.
- Example: human voice.
- A microphone can convert voice into an analog signal.
- Voice can also be sampled and converted into digital data.

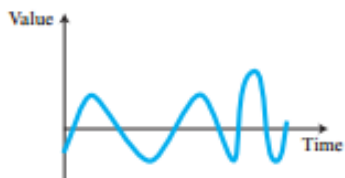
Digital Data

- Discrete in nature.
- Computer memory stores information using 0s and 1s.
- Digital data can be represented by a digital signal.
- It can also be modulated into an analog signal.

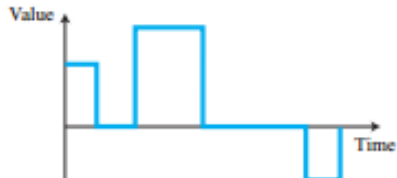
Remember

Data may be **analog or digital**; signals may also be **analog or digital**.

Figure 7.2 *Comparison of analog and digital signals*



a. Analog signal



b. Digital signal

Analog and Digital Signals

Analog Signal

- Continuous waveform.
- Has infinitely many intensity levels.
- Signal value changes continuously with time.

Digital Signal

- Has a limited number of defined values.
- Commonly represented using two levels.
- Example:

1 = positive voltage

0 = zero voltage

Signal Graph

- Vertical axis → signal value / strength.
- Horizontal axis → time.

Analog Signals: Periodic and Nonperiodic

Periodic Analog Signal

A periodic signal completes a pattern within a measurable time and then repeats the same pattern.

- The repeated pattern is called a **cycle**.
- The time required for one cycle is called the **period**.

Nonperiodic (Aperiodic) Signal

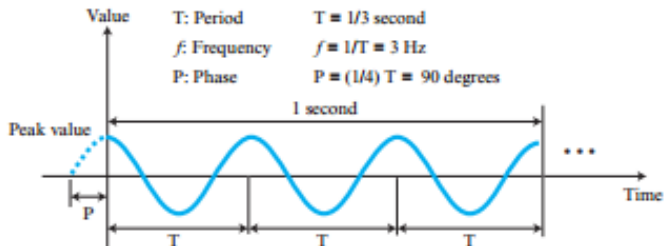
A nonperiodic signal changes continuously without repeating a recognizable pattern or cycle.

In Data Communication

Periodic analog signals are commonly used and can be classified as:

- ① Simple periodic signal
- ② Composite periodic signal

Figure 7.3 A sine wave



Sine Wave – Fundamental Analog Signal

Simple Periodic Signal

The simplest periodic analog signal is a **sine wave**. It cannot be decomposed into simpler periodic signals.

- A sine wave is a smooth, continuously oscillating waveform.
- It is the fundamental building block for more complex analog signals.
- A composite signal can be constructed by combining multiple sine waves.

$$s(t) = A \sin(2\pi ft + \phi)$$

where:

A Peak amplitude

f Frequency

ϕ Phase

A sine wave is the simplest periodic analog signal and is characterized by amplitude, frequency, and phase.

Sine-Wave Parameters

1. Peak Amplitude

The peak amplitude is the absolute value of the highest intensity of the signal.

- It is proportional to the energy carried by the signal.
- Electrical signal amplitude is commonly measured in volts.

2. Period

The period T is the amount of time required to complete one cycle.

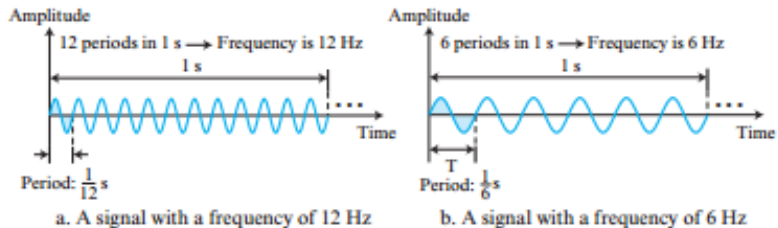
3. Frequency

Frequency is the number of periods completed per second.

$$f = \frac{1}{T}$$

Unit: Hertz (Hz).

Figure 7.4 Wavelength and period



Phase and Wavelength

Phase

Phase describes the position of the waveform relative to time 0.

- It represents the shift of the waveform along the time axis.
- Measured in degrees or radians.
- One complete cycle:

$$360^\circ = 2\pi \text{ radians}$$

- Frequency is independent of the transmission medium.
- Wavelength depends on both frequency and the medium.
- Wavelength is particularly useful when discussing light transmission through optical fiber.

Wavelength

Wavelength λ is the distance a simple signal travels during one period.

$$\lambda = \frac{c}{f}$$

where c is the propagation speed and f is frequency.

Time Domain and Frequency Domain

Time Domain

A time-domain plot shows how the signal amplitude changes with time.

- Horizontal axis: time.
- Vertical axis: amplitude.

Frequency Domain

A frequency-domain plot shows the relationship between amplitude and frequency.

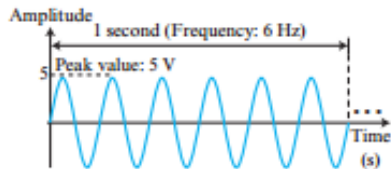
- Horizontal axis: frequency.
- Vertical axis: peak amplitude.

Sine Wave in Frequency Domain

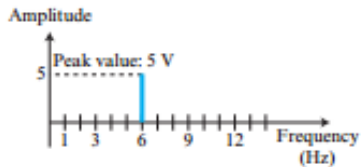
A sine wave appears as a single spike.

- Position of spike $\rightarrow$ frequency.
- Height of spike $\rightarrow$ peak amplitude.

Figure 7.5 *The time-domain and frequency-domain plots of a sine wave*



a. A sine wave in the time domain



b. The same sine wave in the frequency domain

Example 7.1 – Frequency-Domain Representation

Given

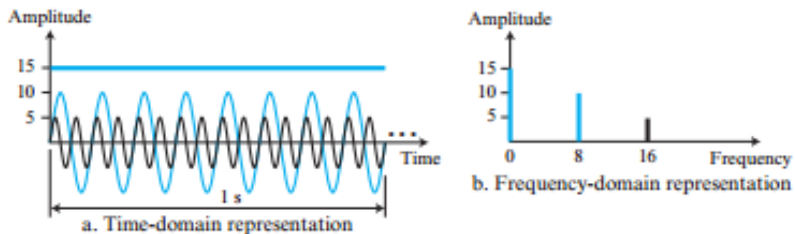
Three sine waves having different amplitudes and frequencies.

- Each sine wave is represented by one spike in the frequency domain.
- The **location** of each spike identifies its frequency.
- The **height** of each spike identifies its peak amplitude.
- Therefore, three sine waves produce three spikes.

Exam Point

One sine wave $\longrightarrow$ one spike in the frequency domain.

Figure 7.6 *The time domain and frequency domain of three sine waves*



Composite Signals and Fourier Analysis

Why Composite Signals?

A single sine wave cannot effectively represent a conversation. It would essentially produce only a simple tone or buzz.

Fourier Analysis

Jean-Baptiste Fourier showed that a composite signal can be represented as a combination of simple sine waves having different:

- amplitudes,
- frequencies,
- phases.

Periodic Composite

Can be decomposed into sine waves with discrete frequencies: $f, 2f, 3f, 4f, \dots$ where f is the fundamental frequency.

Nonperiodic Composite

Can be decomposed into an infinite number of simple sine waves with continuously varying real-valued frequencies.

Bandwidth of a Composite Signal

Definition

Bandwidth is the range of frequencies contained in a composite signal.

$$B = f_{\max} - f_{\min}$$

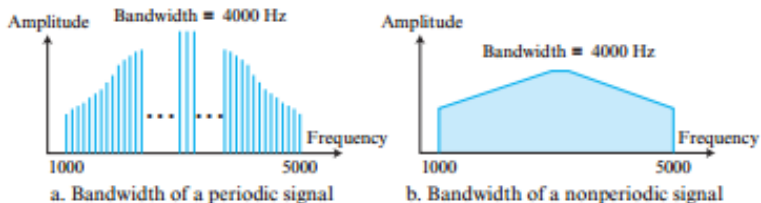
Example

Suppose a composite signal contains frequencies from 1000 Hz to 5000 Hz.

$$B = 5000 - 1000$$

$$B = 4000 \text{ Hz}$$

Figure 7.7 *The bandwidth of periodic and nonperiodic composite signals*



7.7 Two composite signals: one periodic and one nonperiodic. The periodic signal contains discrete frequencies at integral multiples of the fundamental frequency, while the nonperiodic signal contains continuous frequencies over the range.

Digital Signals and Signal Levels

Digital Representation

Information can be represented using a digital signal.

Example:

1 = positive voltage, 0 = zero voltage

Multiple Signal Levels

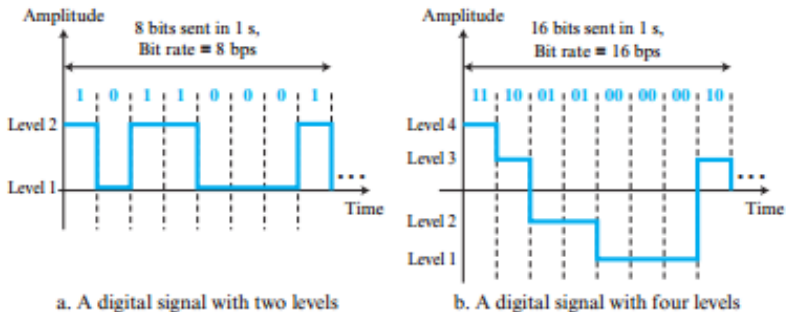
A digital signal does not have to contain only two levels.

If a signal has L levels, the number of bits represented by each signal level is:

$$\log_2 L$$

Signal Levels	Bits/Level	Example
2	$\log_2 2 = 1$	1 bit
4	$\log_2 4 = 2$	2 bits
8	$\log_2 8 = 3$	3 bits

Figure 7.8 Two digital signals: one with two signal levels and the other with four signal levels



Bit Rate – Example 7.2

Bit Rate

For most digital signals, period and frequency are not useful because digital signals are generally nonperiodic.

Therefore we use **bit rate**.

Bit Rate = number of bits transmitted

Unit: bits per second (bps).

Example 7.2

A text download contains 100 pages/minute.

Each page has: 24 lines/page

80 characters/line

8 bits/character

Therefore:

$$100 \times 24 \times 80 \times 8 = 1,536,000 \text{ bps}$$

$$\text{Bit Rate} = 1.536 \text{ Mbps}$$

Bit Length and Digital Signals as Composite Signals

Bit Length

The distance occupied by one bit on the transmission medium is called the **bit length**.

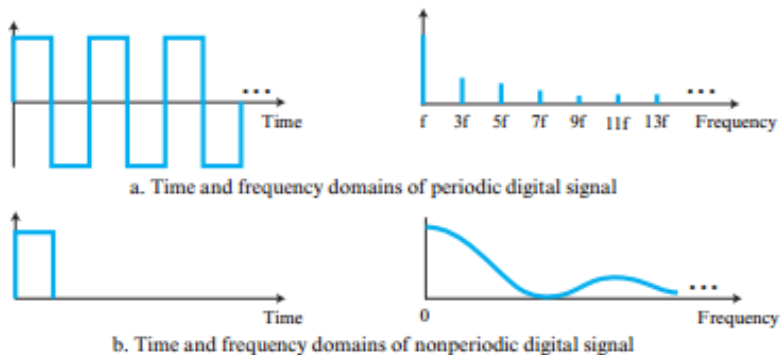
$$\text{Bit length} = \text{Propagation speed} \times \text{Bit duration}$$

Digital Signal as a Composite Analog Signal

Fourier analysis shows that a digital signal can also be viewed as a composite analog signal.

- Its bandwidth is theoretically infinite.
- A vertical transition represents a sudden change: frequency components extend toward infinity.
- A horizontal portion represents no change: frequency zero is present.

Figure 7.9 *The time and frequency domains of periodic and nonperiodic digital signals*



Transmission of Digital Signals – Baseband

Two Approaches

A digital signal can be transmitted using:

- 1 Baseband transmission
- 2 Broadband transmission / modulation

Baseband Transmission

Baseband transmission sends the digital signal directly over the transmission channel without converting it to an analog signal.

- Requires a **low-pass channel**.
- The bandwidth begins at frequency zero.
- A dedicated physical medium may use its complete bandwidth as one channel.
- If several computers share a medium, two stations may communicate at a time using the available channel.

Figure 7.10 *Baseband transmission*



Broadband Transmission

Broadband transmission converts the digital signal into an analog signal before transmission.

- Uses a **band-pass channel**.
- The bandwidth does not begin at zero.
- A low-pass channel can be viewed as a special case of a band-pass channel whose lower frequency is zero.

Figure 7.11 *Bandwidth of a band-pass channel*

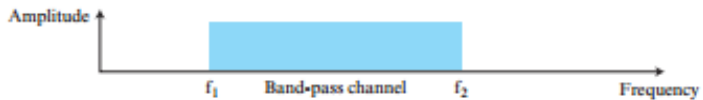
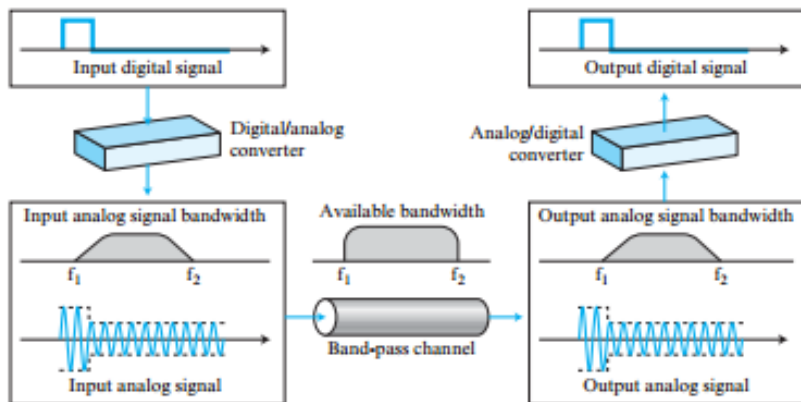


Figure 7.12 Modulation of a digital signal for transmission on a band-pass channel



Example 7.4 – Telephone Subscriber Line

A telephone subscriber line provides a voice channel limited approximately to:

0–4 kHz

If treated as a low-pass/baseband channel, the maximum bit rate would be limited. Instead, digital data is converted into an analog signal and transmitted through the band-pass channel.

- Digital → Analog at sender.
- Analog → Digital at receiver.
- The device performing these functions is a **modem**.

Example 7.5 – Digital Cellular Phone

An analog voice signal is first digitized. The resulting digital data is then converted into a composite analog signal for transmission through the available band-pass channel.

If bandwidth W is shared by 1000 simultaneous caller pairs:

$$\text{Bandwidth per pair} = \frac{W}{1000}$$

Transmission Impairment

Why Does a Signal Change?

A transmission medium is not perfect. The signal received at the destination may differ from the signal transmitted at the source.

Impairment	Meaning
Attenuation	Loss of signal energy
Distortion	Change in signal shape
Noise	Unwanted signal that corrupts the original signal

Attenuation

Signal energy is lost while overcoming resistance in the medium. For example, electrical energy can be converted into heat in a wire. Amplifiers are used to compensate for attenuation.

Attenuation in Decibels – Example 7.6

Decibel

The decibel (dB) measures the relative strength of two signals or the signal strength at two different points.

$$dB = 10 \log_{10} \left(\frac{P_2}{P_1} \right)$$

where P_1 and P_2 are the signal powers at two points.

- Negative dB $\rightarrow$ attenuation / loss.
- Positive dB $\rightarrow$ amplification / gain.

Example 7.6

Suppose:

$$P_2 = 0.5P_1$$

Then:

$$dB = 10 \log_{10}(0.5)$$

$$dB \approx -3 \text{ dB}$$

Thus a loss of approximately 3 dB means that half of the signal power has been lost.

Distortion

Definition

Distortion means that the signal changes its original form or shape during transmission.

- Particularly important for composite signals.
- A composite signal consists of components with different frequencies.
- Different frequency components may propagate at different speeds.
- Therefore, the components may experience different delays.
- These different delays can produce phase differences.
- The receiver reconstructs a composite signal whose shape may differ from the original.

Key Point

Different frequency components arriving with different delays can change the shape of the received composite signal.

Definition

Noise is an unwanted electrical or electromagnetic signal that can corrupt the transmitted signal.

Type	Description
Thermal noise	Random motion of electrons produces an unwanted signal.
Induced noise	Motors and appliances act as transmitting antennas and the medium acts as a receiving antenna.
Crosstalk	One wire interferes with another wire; one behaves like a transmitting antenna and the other like a receiving antenna.
Impulse noise	A high-energy spike of very short duration, such as from power lines or lightning.

Signal-to-Noise Ratio (SNR)

Why SNR?

To determine theoretical data-rate limits, we need to know how strong the desired signal is compared with the noise.

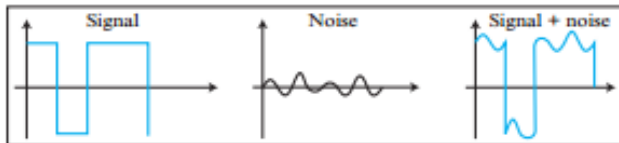
$$SNR = \frac{P_{\text{signal}}}{P_{\text{noise}}}$$

In decibels:

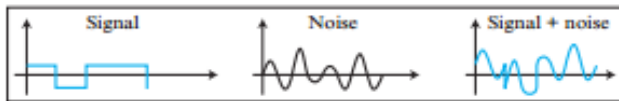
$$SNR_{dB} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right)$$

- High SNR $\rightarrow$ signal is much stronger than noise.
- Low SNR $\rightarrow$ signal is more severely corrupted.
- Average signal power and average noise power are used because both can vary with time.

Figure 7.16 Two cases of SNR: a high SNR and a low SNR



a. High SNR



b. Low SNR

Data Rate Limits

Question

How fast can data be transmitted through a communication channel?

The maximum achievable data rate depends on:

- 1 Available bandwidth
- 2 Number of signal levels
- 3 Quality of the channel / amount of noise

Two Theoretical Limits

Nyquist: Maximum bit rate for a noiseless channel.

Shannon: Maximum theoretical capacity for a noisy channel.

Important

These formulas provide theoretical limits; practical systems must also consider implementation and reliability.

Nyquist Bit Rate

Nyquist Formula – Noiseless Channel

For a noiseless channel:

$$\text{BitRate} = 2B \log_2 L$$

where:

- B = bandwidth in Hz
- L = number of signal levels
- BitRate = bits per second

Effect of Signal Levels

Increasing the number of signal levels theoretically increases the bit rate. However: 2 levels are relatively easy to distinguish, Many levels are harder for the receiver to distinguish and Too many levels can reduce reliability.

Example 7.7 – Nyquist Calculation

Given $B=20,000$ Hz Desired bit rate: BitRate=265,000 bps

Using Nyquist: $265,000 = 2(20,000)\log_2 L$

$$\log_2 L = \frac{265,000}{40,000} = 6.625$$

Therefore:

$$L = 2^{6.625} \approx 98.7$$

Since the number of levels should be a practical integer power of 2:

$$128 \text{ levels} \Rightarrow 2(20,000)\log_2(128) = 280,000 \text{ bps}$$

$$64 \text{ levels} \Rightarrow 2(20,000)\log_2(64) = 240,000 \text{ bps}$$

Conclusion

The desired 265 kbps lies between the values obtainable using 64 and 128 levels.

Shannon Capacity

Noisy Channel

Real communication channels always contain noise.

Shannon's formula gives the theoretical channel capacity:

$$C = B \log_2(1 + SNR)$$

where:

- C = channel capacity in bps
- B = bandwidth in Hz
- SNR = signal-to-noise ratio

Important Difference

Unlike Nyquist's formula, Shannon's formula does not contain the number of signal levels.

No matter how many signal levels are used, the channel capacity cannot be exceeded.

Example 7.8 – Shannon Capacity When $SNR = 0$

Given Suppose:

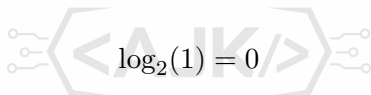
$SNR=0$ Using Shannon's formula:

$$C = B \log_2(1 + SNR)$$

$$C = B \log_2(1 + 0)$$

$$C = B \log_2(1)$$

Since:


$$\log_2(1) = 0$$

we obtain:

$$C = 0$$

Meaning

If there is no useful signal relative to the noise ($SNR = 0$), the theoretical capacity is zero regardless of the available bandwidth.

Example 7.9 – Telephone Line

Given

Telephone line bandwidth:

$$B = 3000 \text{ Hz}$$

SNR:

$$SNR = 3162$$

Shannon capacity:

$$C = 3000 \log_2(1 + 3162)$$

$$C = 3000 \log_2(3163)$$

$$C \approx 34,881 \text{ bps}$$

Example 7.10 – Using Both Limits

Given

$$B=1 \text{ MHz}$$

$$\text{SNR}=63$$

Step 1: Shannon Capacity

$$C=10^6 \log_2(1 + 63)$$

$$C=10^6 \log_2(64)$$

$$C = 6 \text{ Mbps}$$

Thus, the channel cannot theoretically exceed 6 Mbps.

Step 2: Select a Practical Bit Rate Suppose: BitRate=4 Mbps Using

$$\text{Nyquist: } 4 \times 10^6 = 2(10^6) \log_2 L \log_2 L = 2 \boxed{L = 4}$$

Key Relationship

Shannon gives the upper bit-rate limit; Nyquist determines the signal levels required for a selected bit rate.

Performance of a Network

Performance

Performance describes how well a network operates.

Quality of Service (QoS) is discussed in a later chapter.

Metric	Meaning
Bandwidth	Capacity of a channel
Throughput	Actual rate of data transfer
Latency	Time taken for data to travel from source to destination
Bandwidth-delay product	Number of bits that can fill the link
Jitter	Variation in packet delay

Two Meanings of Bandwidth

Bandwidth can be expressed in:

- Hertz (Hz): frequency range.
- Bits per second (bps): data-carrying capacity.

Bandwidth in Hertz and Bits per Second

Bandwidth in Hertz

Bandwidth in Hz is the range of frequencies that a signal or channel can contain or pass.

Example:

0–4 kHz

telephone subscriber line.

Bandwidth in Bits per Second

Bandwidth in bps represents the number of bits transmitted per second.

Example:

100 Mbps

Fast Ethernet.

Relationship

An increase in bandwidth measured in Hz generally allows an increase in bandwidth measured in bps. The exact relationship depends on the transmission method and channel conditions.

Example 7.11

A subscriber line has a bandwidth of 4 kHz and can support up to 56 kbps using a sophisticated modem.

If the telephone company improves the line to 8 kHz:

112 kbps

can be supported under the stated relationship.

Throughput

Definition

Throughput is the **actual rate** at which data is sent through a network.

Suppose a link has:

$$\text{Bandwidth} = 1 \text{ Mbps}$$

but the end devices can process only:

200 kbps

Then:

$$\text{Throughput} = 200 \text{ kbps}$$

Bandwidth vs Throughput

- Bandwidth → potential capacity.
- Throughput → actual achieved rate.
- Normally:

$$\textit{Throughput} < \textit{Bandwidth}$$

Highway Analogy

A highway may be designed for 1000 cars/minute, but congestion may reduce the actual flow to 100 cars/minute.

Designed capacity $\approx$ Bandwidth

Actual flow $\approx$ Throughput

Definition

Latency is the total time required for a message to completely arrive at the destination, from the time the first bit is sent until the final bit arrives.

Delay	Description
Propagation delay	Time required for the signal to travel through the medium.
Transmission delay	Time required to place all message bits onto the link.
Queuing delay	Time spent waiting in queues before transmission.
Processing delay	Time required to examine/process the packet.

$$Latency = d_{propagation} + d_{transmission} + d_{queuing} + d_{processing}$$

Bandwidth-Delay Product

Definition

The bandwidth-delay product (BDP) is the product of bandwidth and delay.

$$BDP = Bandwidth \times Delay$$

It represents the number of bits that can be present on the link at a particular time.

Case 1

Bandwidth = 1 bps

Delay = 5 s

$$BDP = 1 \times 5 = \boxed{5 \text{ bits}}$$

Thus, at most 5 bits can occupy the link.

Case 2

Bandwidth = 5 bps

Delay = 5 s

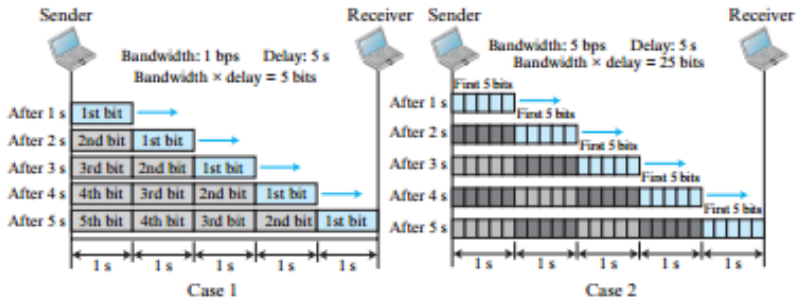
$$BDP = 5 \times 5 = \boxed{25 \text{ bits}}$$

Each bit takes:

$$\frac{1}{5} = 0.20 \text{ s}$$

to occupy the transmission link.

Figure 7.17 *Filling the link with bits for cases 1 and 2*



Bandwidth-Delay Product and Burst Transmission

Using the Link Efficiently

The bandwidth-delay product tells us how many bits can be in transition on a link.

For a full-duplex connection, the textbook explains that a burst size of approximately:


$$2 \times BDP = 2 \times \text{Bandwidth} \times \text{Delay}$$

can be used when an acknowledgment-based protocol is considered.

- Sender transmits a burst of bits.
- Bits travel through the link.
- Receiver sends an acknowledgment.
- The sender waits for the appropriate acknowledgment before sending the next burst.

Example 7.12 – Pipe Analogy

Think of a Communication Link as a Pipe

The textbook uses a pipe analogy to explain the bandwidth-delay product.

Communication Concept	Pipe Analogy
Bandwidth	Cross-sectional area of the pipe
Delay	Length of the pipe
Bandwidth $\times$ Delay	Volume of the pipe
Bits in transit	Data currently filling the pipe

$$BDP = Bandwidth \times Delay$$

Intuition

A high-bandwidth, high-delay link can contain a large amount of data while the data is still travelling toward the destination.

Jitter

Definition

Jitter is the variation in the delay experienced by different packets belonging to the same communication.

Example

Consider three packets:

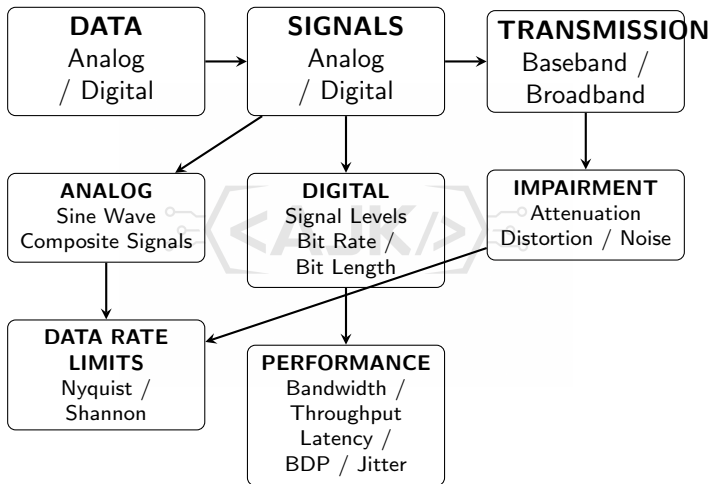
Packet	Delay
1	20 ms
2	45 ms
3	40 ms

The packets do not experience the same delay.

20 ms, 45 ms, 40 ms

This variation is called **jitter**.

Section 7.1 – Complete Concept Map



Key Formulas – Section 7.1

Concept	Formula
Frequency	$f = \frac{1}{T}$
Wavelength	$\lambda = \frac{c}{f}$
Bits per signal level	$\log_2 L$
Bandwidth	$B = f_{\max} - f_{\min}$
Bit length	speed $\times$ bit duration
Attenuation	$dB = 10 \log_{10} \left(\frac{P_2}{P_1} \right)$
SNR	$\frac{P_{\text{signal}}}{P_{\text{noise}}}$
SNR in dB	$10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right)$
Nyquist	$BitRate = 2B \log_2 L$
Shannon	$C = B \log_2 (1 + SNR)$
BDP	$Bandwidth \times Delay$
Latency	$d_p + d_t + d_q + d_{proc}$

Exam-Oriented Summary

- Data can be **analog or digital**.
- Signals can be **analog or digital**.
- Sine wave parameters:

Amplitude, Frequency, Phase

- Frequency and period:

$$f = \frac{1}{T}$$

- Wavelength:

$$\lambda = \frac{c}{f}$$

- Composite-signal bandwidth:

$$B = f_{\max} - f_{\min}$$

- Digital signal with L levels carries:

$$\log_2 L \text{ bits/level}$$

Quick Check – 7.1 Data and Signals

- 1 What is the difference between analog and digital data?
- 2 What are the three parameters of a sine wave?
- 3 What is the relationship between frequency and period?
- 4 What is wavelength?
- 5 How does a sine wave appear in the frequency domain?
- 6 What is a composite signal?
- 7 How is bandwidth calculated?
- 8 How many bits can one signal level represent when $L = 16$?
- 9 What is the difference between baseband and broadband transmission?
- 10 What are the three major transmission impairments?
- 11 What does SNR indicate?
- 12 When should Nyquist's formula be used?
- 13 When should Shannon's formula be used?
- 14 What is the difference between bandwidth and throughput?
- 15 What are the four components of latency?
- 16 What does the bandwidth-delay product represent?
- 17 What is jitter and why is it important for audio/video?